

Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-371;latex-formulas-745;latex-formulas-629;latex-formulas-764;latex-formulas-560
Source: 22 nonblank lines, 2515 chars

p1

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\left|\sum_{\alpha\beta\gamma\delta}\langle\Phi_\alpha\Phi_\beta\Phi_\gamma\Phi_\delta\rangle_c/\sum_{\alpha\beta\gamma\delta}\langle\Phi_\alpha\Phi_\beta\Phi_\gamma\Phi_\delta\rangle=\left|\langle\Phi^4(x)\rangle_c/\langle\Phi^4(x)\rangle\right|\;(n=4)\;,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\int_0^{2\pi}d\sigma X^i(\sigma)\hat{P}_i(\sigma)=x_0^i\hat{p}_0^i+\frac{g_{ij}}{2\sqrt{\alpha'}}\sum_{n=1}^\infty\left\{\left(\chi_n^i\alpha_{-n}^j+\bar{\chi}_n^i\bar{\alpha}_{-n}^j\right)+\left(\bar{\chi}_n^i\alpha_n^j+\chi_n^i\bar{\alpha}_n^j\right)\right\}\;.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\phi_1(x),\left(\phi_2(x),\left(\phi_4(x),\left(\phi_5(x)\right)\right)\right)\right)\rightarrow\left(e^{ \lambda x}\phi_1(x),U\left(e^{-\lambda'x}\phi_2(x),V\left(e^{\lambda''x}\phi_4(x),e^{\lambda'''x}\phi_5(x)\right)\right)\right)$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\frac{-i\langle Ta_{\mathbf{k}}^{\dagger t}(-\mathbf{q})a_{\mathbf{k}'}^{\dagger}(\mathbf{q}')\rangle}{\langle T1\rangle}=\sum_n\exp(\omega_nt)\frac{-i\langle Ta_{\mathbf{k}}^{\dagger n}(-\mathbf{q})a_{\mathbf{k}'}^{\dagger}(\mathbf{q}')\rangle}{\langle T1\rangle}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\bar{G}^{(\zeta)}(\beta)=\frac{(\beta+i\pi)\sinh\left(\frac{\pi\beta}{2\pi+\zeta}\right)}{\Gamma\left(1-\frac{\pi-i\beta}{2\pi+\zeta}\right)\Gamma\left(1-\frac{\pi+i\beta}{2\pi+\zeta}\right)}\;,$$

Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-620;latex-formulas-293;latex-formulas-688;latex-formulas-848;latex-formulas-280
Source: 22 nonblank lines, 2273 chars

p1

Inline and Display Mathematics

Source-backed grouped formula sample

1

Expressions

Expression 1.

The following expression is taken from a source-backed formula corpus.

$$\gamma_{g,9}:|D_{\text{inv}}\rangle\rightarrow\frac{1}{\sqrt{D}}\sum_{\alpha}\sum_{\beta,\gamma}c_{\alpha\beta}\bar{c}_{\gamma\alpha}|d_{\beta}\otimes\bar{d}_{\gamma}\rangle=\frac{1}{\sqrt{D}}\sum_{\alpha}|d_{\alpha}\otimes\bar{d}_{\alpha}\rangle=|D_{\text{inv}}\rangle\;,$$

Expression 2.

The following expression is taken from a source-backed formula corpus.

$$X(\alpha_1,\alpha_2,\alpha_1',\alpha_2')=\exp\left[i\alpha_1q^1\frac{\partial}{\partial q^1}+i\alpha_2q^2\frac{\partial}{\partial q^2}+i\alpha_1'\bar{q}^1\frac{\partial}{\partial \bar{q}^1}+i\alpha_2'\bar{q}^2\frac{\partial}{\partial \bar{q}^2}\right]$$

Expression 3.

The following expression is taken from a source-backed formula corpus.

$$\Gamma^{\mathfrak{t}}_{ij}=a\left(\frac{da}{dt}\right)\gamma_{ij}\,,\quad \Gamma^i_{\mathfrak{t}j}=\frac{1}{a}\left(\frac{da}{dt}\right)\delta^i_j\,,\quad \Gamma^i_{jk}=\frac{\gamma^{\mathfrak{t}l}}{2}(\partial_j\gamma_{lk}+\partial_k\gamma_{lj}-\partial_l\gamma_{jk})\,.$$

Expression 4.

The following expression is taken from a source-backed formula corpus.

$$lA_1+nB_1=lU_l^nStr\left(\ldots\phi^{i'_{2k-1}}\ldots\underbrace{\partial_{a_i}\phi^{j_i}\phi^{i'_{2k}}}_{\phantom{\partial_{a_i}\phi^{j_i}\phi^{i'_{2k}}}}\ldots\right)+nU_l^nStr\left(\ldots\phi^{i'_{2k-1}}\ldots\underbrace{\phi^{i_n}\phi^{i'_{2k}}}_{\phantom{\phi^{i_n}\phi^{i'_{2k}}}}\ldots\partial_{a_i}\phi^{j_i}\ldots\right)$$

Expression 5.

The following expression is taken from a source-backed formula corpus.

$$\Lambda_{\pm}(q_0)=\Lambda(q_0)\pm\frac{2\kappa\left(\frac{2}{\pi}\right)^{\frac{1}{2}}}{[\frac{1}{2}(q_0-1)]!}\left(\frac{1+k}{1-k}\right)^{-\frac{\kappa}{2}}\left(\frac{8\kappa}{1-k^2}\right)^{\frac{1}{2}q_0}\left[1+O\left(\frac{1}{\kappa}\right)\right]$$

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Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-364;latex-formulas-288;latex-formulas-616;latex-formulas-444;latex-formulas-377
Source: 22 nonblank lines, 2149 chars

p1

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1.

The following expression is taken from a source-backed formula corpus.

$$\tilde{\gamma}^a p_{0a} = 0 = \tilde{\gamma}^a f^\mu{}_a p_{0\mu} = (\tilde{\gamma}^m f^\alpha{}_m + \tilde{\gamma}^h f^\alpha{}_h) p_{0\alpha} + (\tilde{\gamma}^m f^\sigma{}_m + \tilde{\gamma}^h f^\sigma{}_h) p_{0\sigma},$$

Expression 2.

The following expression is taken from a source-backed formula corpus.

$$\psi_{-}(\vec{x}) = \frac{1}{\sqrt{2}} \frac{1}{i\partial_{-} - eV_{-}} * \xi(\vec{x}) - \frac{e}{\sqrt{2}} \frac{1}{\Delta_{\perp}} * (J^i_{pzm}(x_{\perp}) - q_i) \alpha^i \frac{1}{i\partial_{-} - eV_{-}} * \psi_{+}(\vec{x}) \;,$$

Expression 3.

The following expression is taken from a source-backed formula corpus.

$$B_{\mu\nu}(t) = \sum_{\rho=0}^N \frac{\eta_{\mu\rho}\eta_{\nu\rho}}{\sqrt{8\omega_{\mu}}} \left[\left(1 + \frac{\omega_{\mu}}{\Omega_{\rho}}\right) \mathrm{e}^{\mathrm{i}\Omega_{\rho}t} + \left(1 - \frac{\omega_{\mu}}{\Omega_{\rho}}\right) \mathrm{e}^{-\mathrm{i}\Omega_{\rho}t} \right] \;.$$

Expression 4.

The following expression is taken from a source-backed formula corpus.

$$\bar{R}^{ab} = R^{ab} - K_c^a \wedge K^{cb} + \frac{1}{2} C_{\cdot c}^{ab} \bar{T}^c + \frac{1}{4} \gamma_J^I \gamma_L^K \theta_I^a \theta_K^b A^J \wedge A^L + i \left(DK^{ab} - \frac{1}{4} \gamma_J^I (\theta_I^a e^b + \theta_I^b e^a) \wedge A^J \right).$$

Expression 5.

The following expression is taken from a source-backed formula corpus.

$$\|\lambda(\phi)\|_{K,m+n} = \sup_{\substack{|p|\leq m+n\\ z\in K}} |D^p(\lambda(\phi))(z)| \;, \qquad D^p = \frac{\partial^{|q|+|r|}}{\partial x_1^{q_1} \cdots \partial x_m^m \partial \theta_1^{r_1} \cdots \partial \theta_n^{r_n}} \;.$$

1

Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-791;latex-formulas-58;latex-formulas-622;latex-formulas-314;latex-formulas-336
Source: 22 nonblank lines, 2561 chars

p1

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$a_{ij}^4 = \frac{1}{g} \sum_{\gamma=1}^r r_{\gamma} \chi_{\gamma}^{R_M \oplus \mathbb{1}^2} \chi_{\gamma}^{(i)} \chi_{\gamma}^{(j)*}$$
$$a_{ij}^6 = \frac{1}{g} \sum_{\gamma=1}^r r_{\gamma} \chi_{\gamma}^{R_M \oplus R_M \oplus \mathbb{1}^2} \chi_{\gamma}^{(i)} \chi_{\gamma}^{(j)*}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{bmatrix} g_2 + w_2^2 h_4 + n_2^2 h_5 & w_2 w_3 h_4 + n_2 n_3 h_5 & w_2 h_4 & n_2 h_5 \\ w_2 w_3 h_4 + n_2 n_3 h_5 & g_3 + w_3^2 h_4 + n_3^2 h_5 & w_3 h_4 & n_3 h_5 \\ w_2 h_4 & w_3 h_4 & h_4 & 0 \\ n_2 h_5 & n_3 h_5 & 0 & h_5 \end{bmatrix},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$D^{++} = u^{+\alpha} \frac{\partial}{\partial u^{-\alpha}}; \quad D^{--} = u^{-\alpha} \frac{\partial}{\partial u^{+\alpha}}$$
$$2D^{++} = [D^0, D^{++}]; \quad -2D^{--} = [D^0, D^{--}]$$
$$D^0 = [D^{++}, D^{--}] = u^{+\alpha} \frac{\partial}{\partial u^{+\alpha}} - u^{-\alpha} \frac{\partial}{\partial u^{-\alpha}}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$A_N = A_1(w_{N-1}) A_{N-1} = \prod_{i=0}^{N-1} A_1(w_{N-i-1})$$
$$D_N = D_1(w_{N-1}) D_{N-1} = \prod_{i=0}^{N-1} D_1(w_{N-i-1})$$
$$\begin{aligned} \text{\textit{beginalign*0.3em}} C_N &= C_1(w_{N-1}) A_{N-1} + D_1(w_{N-1}) C_{N-1} \\ &= \sum_{i=0}^{N-1} D_i(w_{N-1-i}) C_1(w_{N-1-i}) A_{N-1-i}. \end{aligned}$$

p2

Expression 5. The following expression is taken from a source-backed formula corpus.

$$u_i^{35,35}(y,z) = \left\{ \begin{array}{lll} -5b_i^{35,35}(y,z) & if & |5b_i^{35,35}(y,z)| \leq 1, \\ -1 & if & -5b_i^{35,35}(y,z) < -1, \\ 1 & if & 5b_i^{35,35}(y,z) > 1, \end{array} \right\}, \quad i = 1, 2,$$

Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-160;latex-formulas-450;latex-formulas-15;latex-formulas-700;latex-formulas-880
Source: 22 nonblank lines, 2416 chars

p1

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\sum_j q_a^d (|X_j|^2 - |Y_j|^2) \eta_j^+ \bar{\eta}_j^+ = r_a^2 \eta_a^+ \bar{\eta}_a^+ \qquad (a)$$

$$\sum_j q_a^d (X_j \bar{Y}_j) \eta_j^+ \eta_j^+ = 0 \qquad (b)$$

$$\sum_j q_a^d (\bar{X}_j Y_j) \bar{\eta}_j^+ \bar{\eta}_j^+ = 0. \qquad (c)$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$P = \left(\begin{array}{ccccccc} P_{(+++)} & & & & & & \\ & P_{(+-+)} & & & & & \\ & & P_{(+-)} & & & & \\ & & & \ddots & & & \\ & 0 & & & & & \\ & & & & & & P_{(---)} \end{array} \right)$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\begin{array}{cc} \mu_1 & 0 \\ 0 & \mu_2 \end{array}\right) \left(\begin{array}{cc} 0 & d_1 \\ d_2 & 0 \end{array}\right) \left(\begin{array}{cc} \bar{\mu}_1^{-1} & 0 \\ 0 & \bar{\mu}_2^{-1} \end{array}\right) = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right).$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$[q^-, \alpha_0^+] = i \, ,$$

$$[\alpha_n^-, \alpha_m^+] = n \, \delta_{n+m,0} \, ,$$

$$[s_n, s_m] = \left(\frac{k}{2} - 1\right) \, n \, \delta_{n+m,0} \, ,$$

$$[L_n', L_m'] = (n-m) L_{n+m}'$$

Expression 5. The following expression is taken from a source-backed formula corpus.

p2

$$\Sigma + \eta > \frac{1}{M} \sum_{l=1}^M \Big((h^{D_1}(k_0) \phi_l, \phi_l) \|\psi_l\|_{L^2}^2 + (h^{D_2}(K - k_0) \psi_l, \psi_l) \|\phi_l\|_{L^2}^2 \Big)$$

$$\geq (z_1(k_0) + z_2(K - k_0)) \frac{\sum_{l=1}^M \|\phi_l\|_{L^2}^2 \|\psi_l\|_{L^2}^2}{M} = z_1(k_0) + z_2(K - k_0).$$

Dataset: OleehyO/latex-formulas
Source IDs: latex-formulas-732;latex-formulas-991;latex-formulas-647;latex-formulas-216;latex-formulas-719
Source: 22 nonblank lines, 2235 chars

p1

Aligned Mathematical Structures

Source-backed grouped formula sample

1

Expressions

Expression 1.

The following expression is taken from a source-backed formula corpus.

$$= \{\chi^{\mu\nu}\} + \begin{pmatrix} ((p^2 - q^2)a + 2pqb) & (pa + qb) & (pb - qa) & ((p^2 - q^2)a + 2pqb) \\ (pa + qb) & 0 & 0 & (pa + qb) \\ (pb - qa) & 0 & 0 & (pb - qa) \\ ((p^2 - q^2)a + 2pqb) & (pa + qb) & (pb - qa) & ((p^2 - q^2)a + 2pqb) \end{pmatrix}.$$

Expression 2.

The following expression is taken from a source-backed formula corpus.

$$\sigma^* = \begin{cases} \ell + \alpha & 0 < \gamma < \frac{\ell + \alpha}{d} + \frac{1}{p}, \\ \frac{d}{d-1} \left(\ell + \alpha + \frac{1}{p} - \gamma \right) & \frac{\ell + \alpha}{d} + \frac{1}{p} \leq \gamma < \ell + \alpha + \frac{1}{p}, \end{cases}$$

Expression 3.

The following expression is taken from a source-backed formula corpus.

$$z_t = \begin{pmatrix} x_t \\ y_t \end{pmatrix} = U_t(T, D) \begin{pmatrix} x_1' \\ y_1' \end{pmatrix} - D \, U_{t-1}(T, D) \begin{pmatrix} x_0' \\ y_0' \end{pmatrix} + \begin{pmatrix} x^* \\ y^* \end{pmatrix}$$

Expression 4.

The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial t} + \{f, H_T\} \\ &= \frac{\partial f}{\partial t} + \{f, H_T\}_D + \{f, \chi_s\} \left(\Delta^{-1} \right)^{ss'} \{ \chi_{s'}, H_T \} \approx \frac{\partial f}{\partial t} + \{f, H_T\}_D . \end{aligned}$$

Expression 5.

The following expression is taken from a source-backed formula corpus.

$$a^{A*} = a^A \quad b^{A*} = b^A \quad \omega_{AB}{}^* = \omega_{AB} \quad \lambda^* = \lambda \quad T^\dagger = -T$$
$$\bar{\rho}_i = \rho^{i\dagger} \gamma^0 \qquad \bar{\varepsilon}_i = \varepsilon^{i\dagger} \tilde{\gamma}^0$$

1

Dataset: piushorn/arxiv-latex-tables-43k
Source IDs: 2512.04801_table_2;2512.04801_table_3
Source: 32 nonblank lines, 8113 chars

p1

Simple Tables	
Source-backed grouped table sample	
1 Tables	
These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.	
1	

p3

Table 2: Source table 2: 2512.04801_table_3									
qidx	Frequency (GHz)	T1 [ps]	T2 [ps]	Residual Error (%)	$\sqrt{\chi^2}$ (%)	ECR T (%)	ECR L (%)	ECR - (%)	ECR -> (%)
q0	4.80	286.2	286.2	1.0	0.05	0.64	0.6		
q1	4.92	286.3	286.2	1.0	0.05	0.64	0.6		
q2	5.0	277.4	275.4	1.2	0.05		0.55		
q3	4.97	386.4	325.9	2.02	0.03	1.24	0.92		
q4	4.84	386.5	386.5	1.04	0.02	1.75	1.12		
q5	4.81	385.9	342.2	1.1	0.02			0.65	
q6	4.93	389.9	379.2	1.06	0.02			0.5	
q7	4.89	361.4	401.6	31.25	0.02		0.57	1.06	
q8	4.97	287.7	321.1	3.79	0.05			0.65	
q9	4.77	281.2	327.5	3.89	0.05			1.28	
q10	4.79	381.9	327.9	22.3	0.05				
q11	4.84	332.7	335.2	1.62	0.02			0.96	1.43
q12	5.04	271.2	380.7	1.44	0.02				0.54
q13	4.82	384.2	331.9	0.95	0.02				
q14	4.96	229.9	312.5	1.35	0.02			0.59	0.6
q15	4.89	284.5	316.7	0.97	0.02				
q16	5.11	224.1	314.9	4.06	0.02			0.93	0.96
q17	4.75	281.3	316.6	2.89	0.03			0.89	0.59
q18	5.12	312.2	318.8	1.05	0.02				
q19	4.9	345.3	345.9	7.07	0.02				
q20	4.79	281.2	386.7	2.07	0.02				
q21	5.06	282.4	361.4	4.17	0.03				
q22	4.96	302.9	257.7	1.29	0.02	0.56			
q23	5.03	256.8	310.1	1.07	0.04	1.09	1.16		
q24	5.03	259.9	406.3	6.71	0.02			0.98	
q25	4.84	380.1	401.3	1.12	0.05				
q26	5.04	256.3	400.9	5.42	0.05				3.72
q27	4.64	387.2	282.2	1.25	0.02				
q28	4.66	384.9	321.9	6.96	0.05			1.15	0.56
q29	5.02	281.2	316.8	5.06	0.02	0.75		0.93	
q30	4.82	341.2	315.7	1.54	0.1				1.14
q31	4.79	314.9	401.9	1.06	0.02			0.86	0.63
q32	4.96	289.4	312.5	0.96	0.03				
q33	5.1	259.4	322.4	1.09	0.02	0.44		0.63	0.57
q34	4.8	318.7	316.5	4.11	0.04				1.04
q35	4.89	319.4	316.6	10.03	1.39				3.0
q36	5.1	314.1	317.7	0.95	0.05				0.96
q37	4.86	285.8	311.2	1.06	0.04				
q38	4.91	289.9	344.9	1.07	0.05				
q39	4.91	346.2	379.7	1.07	0.04	0.55			
q40	4.91	381.4	317.4	0.55	0.02			1.6	0.57
q41	4.96	312.8	316.1	1.07	0.31	1.4	10.61		
q42	4.98	385.3	315.7	1.1	0.02	1.14			0.65
q43	4.84	225.3	319.9	1.1	0.03				0.49
q44	4.75	380.2	74.6	4.94	0.03				
q45	4.82	284.1	316.7	0.46	0.02			0.59	0.75
q46	4.86	315.2	311.1	1.46	0.11	1.28			
q47	5.06	279.4	382.1	0.67	0.03			0.6	
q48	4.91	349.9	225.9	0.92	0.02			0.96	3.28

2

Table 1: Source table 1: 2512.04801_table_2									
qidx	Frequency (GHz)	T1 [ps]	T2 [ps]	Residual Error (%)	$\sqrt{\chi^2}$ (%)	ECR T (%)	ECR L (%)	ECR - (%)	ECR -> (%)
q0	4.72	281.7	381.1	2.07	0.04				
q1	4.82	381.1	286.5	0.95	0.04			0.44	
q2	4.81	286.8	343.5	0.97	0.06			1.01	
q3	4.85	318.9	389.9	1.9	0.02			0.94	
q4	4.82	351.8	289.9	1.41	0.02			0.87	0.41
q5	4.73	315.7	233.9	0.75	0.03				0.71
q6	4.88	415.7	50.1	0.02	0.02			0.61	0.77
q7	4.97	385.2	388.4	2.65	0.04				0.71
q8	4.9	384.7	222.9	0.9	0.02				0.72
q9	4.88	384.2	375.7	0.79	0.02				
q10	4.85	318.1	286.5	1.05	0.04			0.44	
q11	4.97	344.9	285.1	13.05	0.01				0.68
q12	4.84	386.7	388.4	1.26	0.02			0.45	
q13	5.01	282.3	389.9	1.25	0.03			0.47	0.56
q14	4.8	386.1	314.5	1.56	0.02	0.45			
q15	4.85	283.3	51.8	1.12	0.03			0.95	
q16	4.86	336.1	41.1	1.48	0.03	0.48		1.57	
q17	4.82	335.7	386.7	1.28	0.02			0.96	
q18	4.79	279.8	388.6	0.17	0.02				
q19	4.75	314.1	381.1	8.46	0.07				0.51
q20	4.86	362.8	381.2	1.32	0.07			3.38	2.03
q21	4.97	394.2	319.7	0.72	0.02				
q22	5.04	211.1	347.9	1.46	0.06				1.08
q23	4.84	311.2	282.3	1.29	0.05				1.38
q24	5.1	233.8	178.1	20.19	0.02			0.71	1.09
q25	4.95	289.4	233.3	1.53	0.02			0.51	
q26	4.85	385.1	1.9	0.02	0.07			1.2	
q27	4.75	383.7	383.2	3.45	0.02			3.35	
q28	4.89	318.1	389.9	3.15	0.03				1.26
q29	4.83	350.3	330.4	1.98	0.02			0.91	1.3
q30	4.71	316.2	316.1	0.95	0.02				
q31	4.8	237.4	431.4	0.7	0.02				0.53
q32	4.81	250.2	320.4	0.94	0.02			0.54	0.7
q33	4.97	393.3	311.1	0.9	0.02			0.8	
q34	4.88	286.1	316.3	2.05	0.03			1.2	
q35	4.91	194.4	315.1	1.29	0.02			0.92	
q36	4.81	316.3	316.4	1.89	0.03			0.3	
q37	4.84	381.2	381.6	0.97	0.12				0.96
q38	4.78	281.1	354.6	1.42	0.02				
q39	4.82	222.2	316.6	2.52	0.03				1.59
q40	4.87	224.4	314.4	3.92	0.02				1.14
q41	4.84	388.8	286.4	4.71	0.02			0.57	0.45
q42	5.06	351.2	311.1	0.99	0.02				1.23
q43	4.83	356.1	365.1	5.1	0.06				3.02
q44	4.95	319.1	318	2.57	0.03				0.9
q45	5.0	387.4	271.9	0.1	0.14				
q46	4.85	386.1	315.3	1.22	0.04			1.55	3.05
q47	4.77	384.8	386.8	0.92	0.02				
q48	4.84	270.7	311.6	1.24	0.02			0.48	1.46
q49	4.7	264.3	318.8	0.79	0.03				1.05
q50	4.78	337.8	44.9	1.32	0.04			1.08	1.07

2

Dataset: piushorn/arxiv-latex-tables-43k
Source IDs: 2512.04801_table_4;2512.02677_table_2
Source: 55 nonblank lines, 4119 chars

p1

Simple Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.04801_table_4									
qubit	Frequency (GHz)	T1 (μ s)	T2 (μ s)	Readout Error (%)	$\sqrt{\chi}$ (%)	ECR $\uparrow$ (%)	ECR $\downarrow$ (%)	ECR $\leftarrow$ (%)	ECR $\rightarrow$ (%)
q_{101}	5.07	181.4	224.5	2.55	0.02			1.07	0.63
q_{102}	4.81	213.7	160.3	4.16	0.02				0.87
q_{103}	4.93	144.8	86.4	0.72	0.03				
q_{104}	4.77	196.3	176.0	1.29	0.02			0.71	
q_{105}	4.98	172.6	98.8	0.76	0.03			0.56	0.69
q_{106}	4.88	219.4	186.2	1.68	0.03				
q_{107}	4.92	181.6	119.6	2.12	0.02			0.62	
q_{108}	5.06	163.6	160.9	2.15	0.03		0.73	0.68	
q_{109}	4.98	184.1	227.2	2.53	0.03	0.81			
q_{110}	4.83	202.2	276.2	1.19	0.03		0.86		
q_{111}	4.85	259.1	294.9	1.22	0.02	0.52			
q_{112}	5.0	176.1	67.2	0.47	0.02		0.79		
q_{113}	5.03	206.6	209.3	0.83	0.02				1.09
q_{114}	4.82	185.2	61.0	13.86	0.02	0.61			0.98
q_{115}	5.03	176.0	92.6	0.84	0.02				
q_{116}	4.91	205.8	191.0	0.77	0.01			0.57	0.56
q_{117}	4.83	247.6	213.9	1.56	0.03				0.58
q_{118}	4.73	241.4	116.4	0.76	0.02				0.99
q_{119}	4.8	196.0	122.4	32.95	0.07				
q_{120}	4.84	184.5	154.6	17.81	0.03			1.58	
q_{121}	4.97	234.4	143.5	1.84	0.05			2.23	
q_{122}	4.94	224.1	130.2	2.21	0.05	0.75		1.36	0.84
q_{123}	5.03	145.8	126.9	1.64	0.02				
q_{124}	4.95	214.2	194.8	0.82	0.03			0.59	
q_{125}	4.86	258.3	223.5	1.05	0.01			0.59	0.58

p2

Table 2: Source table 2: 2512.02677_table_2			
Components	Boolean	Propositional	Arithmetic
constants $\mathcal{C}$	1 0	0000	000
		0001	001
		0010	002
		...	...
		1111	999
functions $\mathcal{F}$	$+^2$	$+^2$	$+^2$
	$*^2$	$*^2$	$-^2$
	$-^1$	$-^1$	$*^2$
	$>^2$		
base-cases $\mathcal{B}$	11+	11000101 >	453857+
	10*	10100011+	053007*
	0−	10001110*	403857−
	...	...	...
recursion depth δ	$\delta(11 + 10 * *) = 2$	$\delta(110011001000 + *) = 2$	$\delta(222111 + 003*) = 2$
	$\delta(110 * *) = 2$	$\delta(11001011 >) = 1$	$\delta(333003*) = 1$
	$\delta(10*) = 1$	...	...
	...		
expr. length ℓ	$\ell(11 + 10 * *) = 3$	$\ell(110011001000 + *) = 2$	$\ell(222111 + 003*) = 2$
	$\ell(110 * *) = 2$	$\ell(11001011 >) = 1$	$\ell(333003*) = 1$
	$\ell(10*) = 1$	$\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}$	
	...	...	...

Dataset: piushorn/arxiv-latex-tables-43k
Source IDs: 2512.01992_table_15;2512.02839_table_2
Source: 172 nonblank lines, 7761 chars

p1

Simple Tables			
Source-backed grouped table sample			
1 Tables			
These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.			
1			

p3

Table 2: Source table 2: 2512.02839_table_2			
Source Name	State	alpha (LP)	beta
PKS 0736+017 $\bar{\alpha} = 2.33 \pm 0.26$ $\bar{\beta} = 0.46 \pm 0.22$	Low	2.67±0.13	0.18±0.12
	High	2.38±0.04	0.13±0.04
	Full	2.33±0.01	0.116±0.009
	VHE	2.38±0.05	-0.02±0.03
PKS 1510-089 $\bar{\alpha} = 2.39 \pm 0.12$ $\bar{\beta} = 0.07 \pm 0.07$	Low+VHE	2.18± 0.05	-0.02± 0.02
	High	2.48±0.02	0.09±0.01
	Full	2.398±0.004	0.059±0.003
	VHE	2.38±0.02	0.04±0.01
4C+21.35 $\bar{\alpha} = 2.46 \pm 0.16$ $\bar{\beta} = 0.14 \pm 0.14$	Low	2.65±0.08	-0.06± 0.03
	High	2.10±0.01	0.079 ±0.008
	Full	2.303± 0.007	0.054±0.004
	VHE	2.12±0.05	0.08±0.02
3C 279 $\bar{\alpha} = 2.36 \pm 0.09$ $\bar{\beta} = 0.09 \pm 0.07$	Low	2.29± 0.09	0.06±0.05
	High	2.024± 0.009	0.088±0.006
	Full	2.222±0.004	0.082±0.003
	VHE	2.24±0.02	0.06±0.01
TON 0599 $\bar{\alpha} = 2.25 \pm 0.18$ $\bar{\beta} = 0.15 \pm 0.14$	Low	2.32±0.26	0.26±0.19
	High	1.83±0.02	0.079±0.009
	Full	2.071±0.007	0.069±0.004
	VHE	1.95±0.03	0.06±0.02
PKS 1441+25 $\bar{\alpha} = 2.14 \pm 0.20$ $\bar{\beta} = 0.28 \pm 0.21$	Low	2.16±0.13	0.018±0.06
	High	1.89±0.029	0.10±0.02
	Full	2.08±0.02	0.08±0.01
	VHE	1.83±0.06	0.13±0.04
S3 0218+35 $\bar{\alpha} = 2.28 \pm 0.21$ $\bar{\beta} = 0.20 \pm 0.16$	Low	2.09±0.14	0.07±0.07
	High	2.32±0.02	0.12±0.01
	Full	2.27±0.01	0.056±0.007
	VHE	1.62±0.11	0.05±0.04

p2

Table 1: Source table 1: 2512.01992_table_15		
Model	Avg. Tokens/Move	Avg. Cost / Game
o3 (low)	1927.5	\$8.1653
o4-nimi (high)	5695.2	\$2.7146
o3 (medium)	5040.3	\$7.3626
o1 (medium)	3309.1	\$19.5655
o4-nimi (medium)	2155.6	\$1.1091
o1 (low)	1638.9	\$13.4843
o4-nimi (low)	680.23	\$0.5273
o3-nimi (medium)	2337.8	\$1.6658
o1-preview	2660.1	\$22.5618
Claude 3.7 Sonnet Thinking	671.33	\$2.0754
Claude 3.7 Sonnet	262.81	\$0.8993
GPT-4.32K	6.06	\$2.2266
Claude 3.5 Sonnet v2	91.15	\$0.5590
Qwen2.5-Max	6.06	\$0.1336
GPT-4 Turbo	6.06	\$0.8482
GPT-4o (2024-11-20)	51.59	\$0.3165
GPT-4.1	18.94	\$0.1976
GPT-4.5	8.03	\$6.4834
GPT-4	8.21	\$1.8986
Claude 3.5 Haiku	67.72	\$0.0465
GPT-4o (2024-08-06)	7.7	\$0.2081
Claude 3.5 Sonnet	88.13	\$0.5658
Genini 2.5 Pro Preview	434.93	\$0.5570
GPT-4o (2024-05-13)	31.34	\$0.2669
o3-nimi (low)	669.8	\$0.4827
Deepseek-R1	4585	\$0.9375
GPT-4.1 Mini	8.2	\$0.0172
GPT-4o Mini	104.64	\$0.0215
Llama-3-70B-Instruct	41.61	\$0.0205
Genini 2.0 Flash	93.77	\$0.0147
Grok-2	66.23	\$0.1904
Genini 1.5 Flash	19.91	\$0.0034
Gemma 2 27B	55.04	\$0.0199
Gemma 2 9B (8bit)	58.12	\$0.0014
DeepSeek-V3 (0324)	410.71	\$0.0470
Llama-3.3-70B	102.98	\$0.0140
Qwen Plus	440.41	\$0.0728
Qwen2.5-72B-Instruct	219.47	\$0.0110
Genini 2.0 Flash (exp)	168.15	\$0.0115
Llama-3.1-8B	162.1	\$0.0009